Machine Code Recognition

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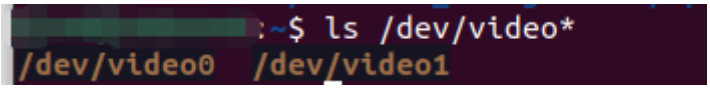
Hardware Connection

Plug the USB camera into the motherboard's USB port. Open the system terminal and enter the following command to check the camera's device number. If /dev/video0 appears, it's normal.

```
ls /dev/video*
```



Install Driver Libraries

```
sudo apt install libzbar-dev
sudo pip3 install pyzbar --break-system-packages
sudo pip3 install pupil-apriltags --break-system-packages
```

例程代码

```
from pupil_apriltags import Detector
import cv2 as cv
from pyzbar.pyzbar import decode
import time
import numpy as np
from PIL import Image, ImageDraw, ImageFont

# Create a detector instance (parameters can be specified)
at_detector = Detector(
    families="tag36h11",  # Specify Tag Family
    nthreads=1,
    quad_decimate=1.0,
    quad_sigma=0.0,
    refine_edges=1,
    decode_sharpening=0.25,
    debug=0
)


def decodeDisplay(image):
    gray = cv.cvtColor(image, cv.COLOR_BGR2GRAY)
```

```
    # Perform Detection
    results = at_detector.detect(gray)
    for tag in results:
        # 提取机器码的边界框的位置
        # Extract the position of the boundary box of the apriltag
        (c1, c2, c3, c4) = tag.corners
        x = int(c4[0])
        y = int(c4[1])
        x1 = int(c2[0])
        y1 = int(c2[1])
        cv.rectangle(image, (x, y), (x1, y1), (225, 0, 0), 5)

        # 绘出图像上数据和类型
        # Draw the data and type on the image
        pilimg = Image.fromarray(image)
        # 创建画笔
        # create brush
        draw = ImageDraw.Draw(pilimg)
        # # 参数1：打印坐标，参数2：文本，参数3：字体颜色
        # Parameter 1: print coordinates, parameter 2: text, parameter 3: font
color
        text = str(tag.tag_family).lstrip('b\'').rstrip('\'') + ' - ' +
str(tag.tag_id)
        draw.text((x, y - 40), text, fill=(255, 0, 0))

        # 绘出标签码的中心点
        # Draw the center of the apriltag
        tag_center = tag.center
        radius = 5
        draw.ellipse([(tag_center[0]-radius, tag_center[1]-radius),
(tag_center[0]+radius, tag_center[1]+radius)], fill="red")

        # # PIL图片转cv2 图片
        # PIL picture to CV2 picture
        image = cv.cvtColor(np.array(pilimg), cv.COLOR_RGB2BGR)
        # 向终端打印信息
        # Print information to terminal
        print(f"Tag Family: {tag.tag_family}")
        print(f"Tag ID: {tag.tag_id}")
        print(f"Center: {tag.center}")
        print(f"Corners: {tag.corners}")
        print("---")
    return image

if __name__ == '__main__':
    capture = cv.VideoCapture(0)
    cv_edition = cv.__version__
    if cv_edition[0] == '3': capture.set(cv.CAP_PROP_FOURCC,
cv.VideoWriter_fourcc(*'XVID'))
    else: capture.set(cv.CAP_PROP_FOURCC, cv.VideoWriter.fourcc('M', 'J', 'P',
'G'))
    capture.set(cv.CAP_PROP_FRAME_WIDTH, 640)
    capture.set(cv.CAP_PROP_FRAME_HEIGHT, 480)
    print ("capture get FPS : ",capture.get(cv.CAP_PROP_FPS))
    while capture.isOpened():
        start = time.time()
        ret, frame = capture.read()
        action = cv.waitKey(10) & 0xFF
```

```
        frame = decodeDisplay(frame)
        end = time.time()
        fps = 1 / (end - start)
        text = "FPS : " + str(int(fps))
        cv.putText(frame, text, (30, 30), cv.FONT_HERSHEY_SIMPLEX, 0.6, (100,
200, 200), 1)
        cv.imshow('frame', frame)
        if action == ord('q') or action == 113: break
    capture.release()
    cv.destroyAllWindows()
```

Running the Program

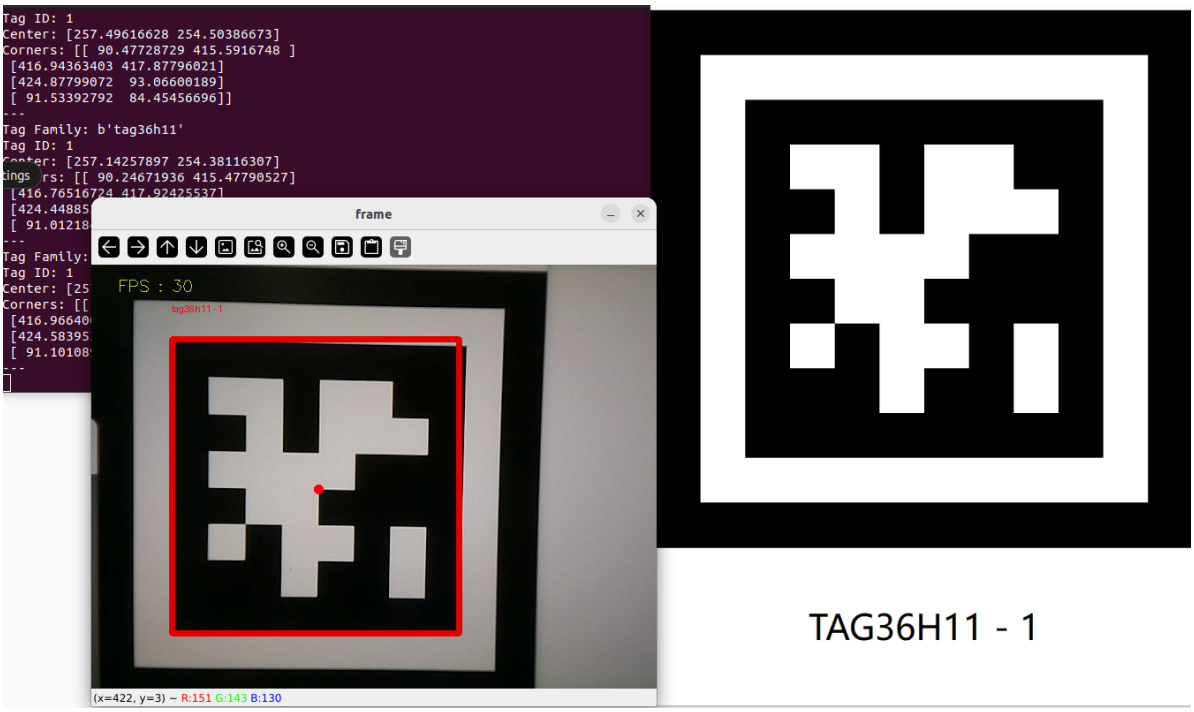
Open a desktop terminal and execute the program:

```
cd ~/content_recognition/apriltag/
python apriltag_parsing_usb.py
```

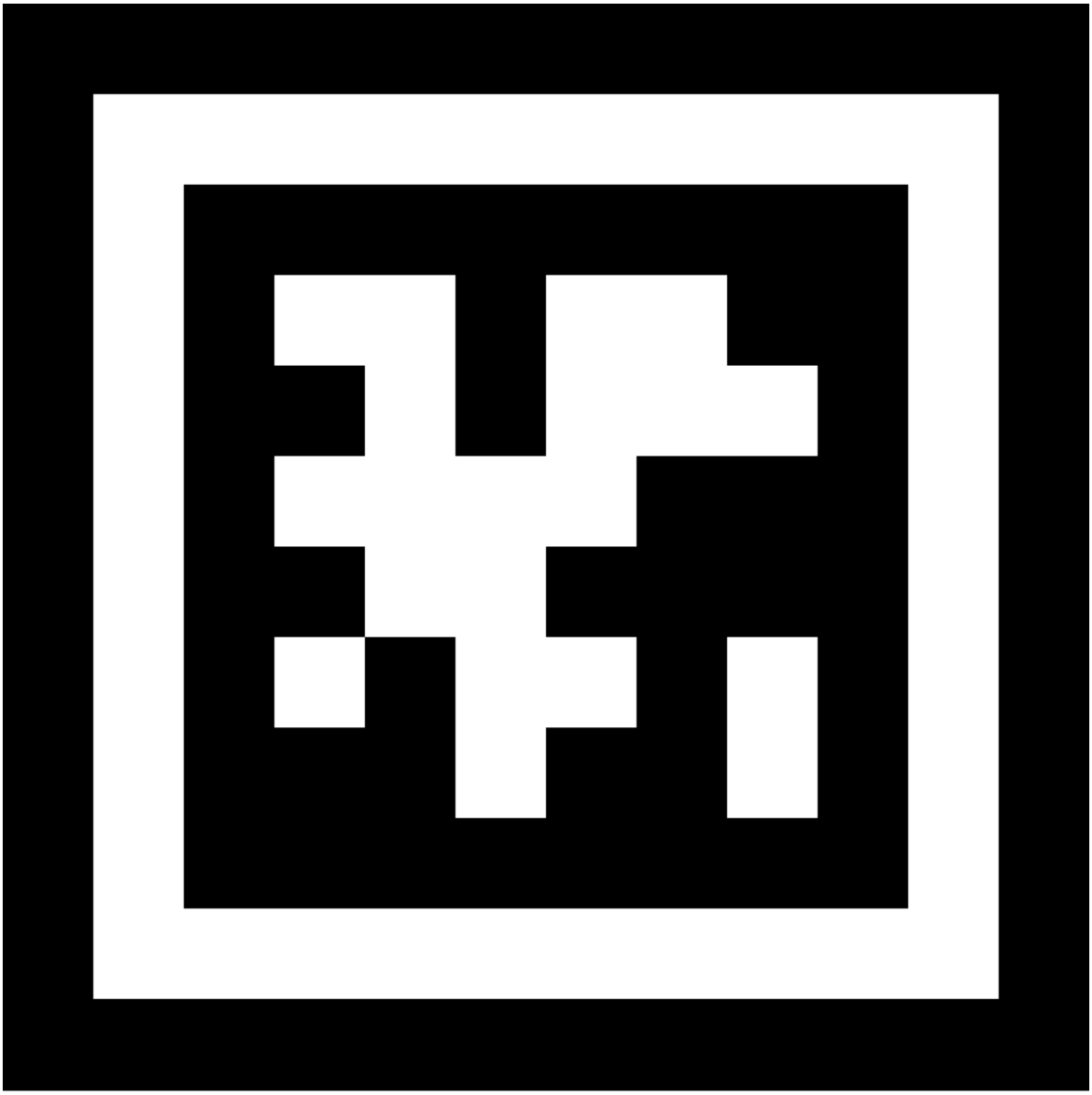
Note: Since the program requires a display window to show the video feed, do not run it via a remote SSH terminal. Please run the program using the native desktop environment or a VNC remote desktop session instead.

Program Output

Upon execution, the program automatically activates the USB camera. Simply point the camera at an AprilTag; the program will then automatically frame the tag in the video feed and print its associated information to the terminal.



Sample AprilTag Image:



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